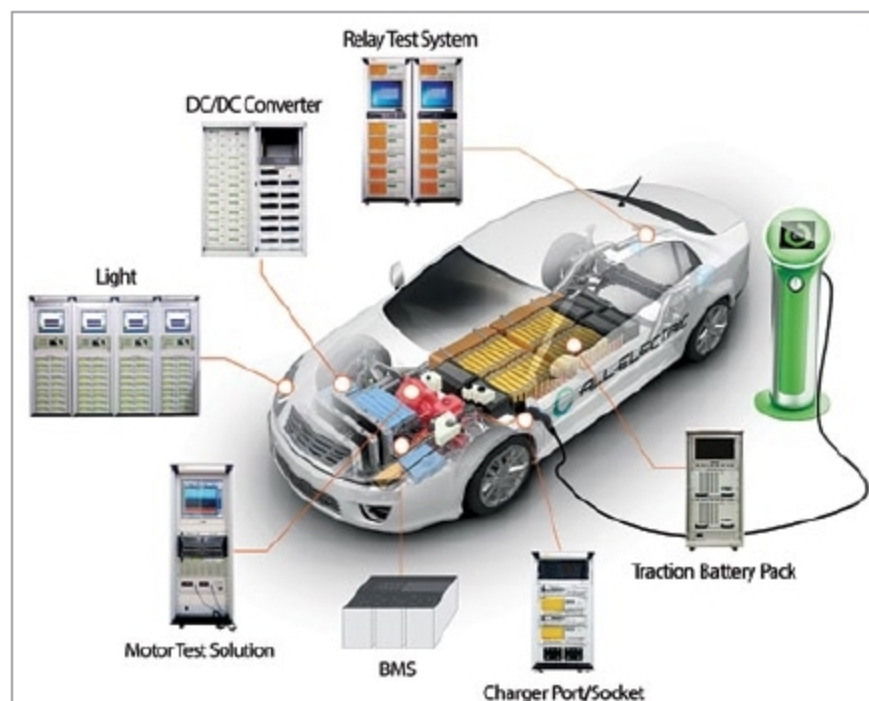




A battery test system (Credit: www.odacore.com)

nectors, as well as charging stations. Many laboratories are replicating a variety of environmental conditions, such as temperature extremes, vibrations, shocks and impact, and simulating hazards that may arise during vehicle transportation or operation to evaluate performance.

EV battery test system

Battery testers should be regenerative, high-precision systems specifically designed for cell-, module- and pack-level battery testing. From charge/discharge to drive cycle simulation, accuracy in measurements ensures exact and reliable testing for battery capacity, performance, production and qualification. The rapidly-emerging EV, hybrid EV (HEV) and plug-in HEV (PHEV) markets are poised for continued growth and change.

Hybrid and electric batteries test conditions include cell-, module- and pack-level testing, cycle life hybrid pulse power characterisation (HPCC), safety certification testing, overcharge, short-circuit, mechanical shock, thermal shock, vibration, temperature, humidity, failure analysis, battery management solution (BMS) simulation and monitoring, connector/service disconnect cycling and durability, battery safety, battery performance, battery fire and abuse, engage/disengage force, maximum current and mechanical overstress.

There are three main types of EV charging points, each defined by power output and, hence, how quickly

ing for charge and discharge with freely programmable set values for current, voltage, power or resistance in test programs. Due to the specific requirements of fast load changes and simulation of driving cycles, this type of testing uses IGBT technology. High-speed transition from charge to discharge, and vice versa, make this technology ideal for EV and HEV battery testing. A portable data logger used to record drive cycles during a real test drive provides reproducible test results.

BMS testing. Introduction of new storage technologies and interconnection of multiple energy storage cells to form modules require an intelligent BMS. It predicts safety, control and regulation functions by measuring various parameters such as voltage, current, temperature or state of charge. It is also responsible for thermal management, energy management, cell balancing and performance, which is a great challenge in BMS development and validation.

For reproducible testing and BMS optimisation, modern energy storage devices have to meet high-quality and market demands for power and energy density, safety and durability. This requires comprehensive tests to be carried out to ensure successful battery production.

EV motor testing

To ensure good quality of the motor, dynamic power of an EV needs to be tested with impulse/surge testers and

these can charge an EV. Slow charge (up to 3kW) can fully charge an EV in around six to eight hours. Fast charging (7kW - 22kW) time is two to four hours. Rapid charge (up to 50kW) typically takes under 30 minutes. Most EVs come with multiple cables to allow connections with a wide range of EV charging points.

High-speed transition testing.

This includes test-

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